Urban Structure and Commuting Patterns in the Age of the Streetcar: Worcester in 1918

Sarah Philbrick ’15 (Sponsor: Prof. John Brown)

Abstract

The ability of a city to thrive has always depended upon its urban structure, the outcome of tradeoffs both firms and workers make in deciding where to locate. The Mills-Muth monocentric city model has been the urban economist’s answer to assess the impact of the tradeoff for households on urban structure. This study uses historical commuting data from Worcester, MA to test the validity of the monocentric city model in the age of the streetcar. This study finds that in choosing where to live, distance to work and housing density are a joint decision. For immigrants, living near other immigrants with the same ethnic background is the most influential driver in their decision.

Introduction

“What determines where people choose to live?” is a fundamental question in urban economics. Because of this, any more information on whether a tradeoff between commuting costs and the amount of housing consumed existed when people had a choice between a fast mode (the street car being about 14 miles an hour) and a slow mode (walking being about 3 miles an hour) is useful in understanding tradeoffs in cities today.

Economic Framework

1918 Worcester, MA serves as the basis for this study because of its likelihood to be a monocentric city. It has a high propensity for income sorting, a highly industrial center, and rapid growth. While 1918 Worcester has many of the characteristics of a monocentric city, recent literature shows the classical Mills-Muth monocentric city model cannot explain the spatial structure of most cities without accounting for exogenous factors. The relationships are thought to be as follows:

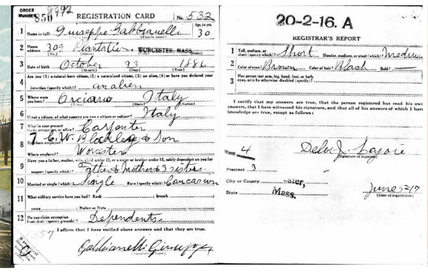
Distance=f*(income, job security, ethnicity, immigration status, home ownership, neighborhood wealth)*

Space= g*(income, ethnicity, immigration status, home ownership, neighborhood wealth)*

Data and Variable Creation

A random sample of 1,024 Worcester City WWI draft cards provided by the National Archives and Records Administration is used for this research. 784 individuals in the sample have both a valid home and valid firm address within the city of Worcester and are used in this study. The Euclidian distance between the two points is taken to determine a commuting distance for each individual.

Occupations are coded using the 1950 Department of Commerce Index to Occupations. The most popular are “Laborers” (12.5%), “Operatives and kindred workers” (11.1%), and “Machinists” (7.2%). Each individual is given a skill level 1-6 based on these coded occupations



51.8% of the sample are immigrants, naturalized, or have a naturalized father, while 48.2% of the sample are natural born citizens of the United States. A dominant ethnicity variable is created for individuals who are of an ethnicity and live in a ward where their ethnicity is most prevalent in Worcester. Another variable, very close to church, indicates a foreign born individual lives within .25km of a church of their ethnicity.

A density measure is created using the number of neighbors in the sample within .25km of an individual’s household.

Empirical Models

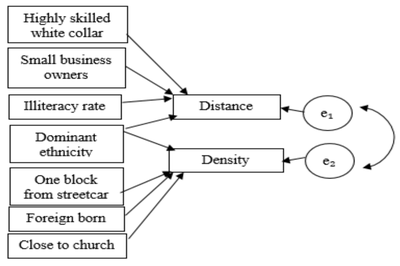
Three empirical models are used along with various spatial analysis techniques.

The first empirical model assumes that density can be explained as an ordered categorical variable and housing density and commuting distance are decisions that can be made separately in choosing where to live. This is represented by:

$$\textbf{\textit{Indistance}}=\beta_0 + \beta_1 skill1 + \beta_2 skill6 + \beta_3 dominant_ethnicity + \beta_4 illiteracyrate + u$$

$$\textbf{\textit{logit(density}}_j\textbf{\textit{)}}= \alpha_j - \beta_1 skill2 - \beta_2 skill3 - \beta_3 skill4 - \beta_4 skill5 - \beta_5 skill6 - \beta_6 foreignborn - \beta_7 domimant_ethnicity - \beta_8 close_to_church + u$$

The second empirical model assumes both housing density and commuting distance can be modeled using an OLS regression with correlated error terms (seemingly unrelated regression) A structural equation model is used to fit the model to:



The third empirical takes into account the fact that density and distance are a joint decision in choosing a place to live. Distance an density are made into a matrix of choices (categorical) and a multinomial logistic regression is used with the equation:

$$\textbf{\textit{Logit(choice}}_j\textbf{\textit{)}}= \beta_0 + \beta_1 skilled_bluecollar + \beta_2 unskilled_bluecollar + \beta_3 foreign_born + \beta_4 dominant_ethnicity + u$$

After these models were run, a grouping analysis in ArcGIS was run to see if there were spatial clusters of groups that were previously unseen. Standard deviational ellipses are created to show the directional distribution of different groups.

Results

The first model showed foreign born individuals and those who live in a census ward where their ethnicity is dominant have a high probability of living in a dense area at 37% and 53%, respectively. This model also showed being a small business owner or highly skilled white collar worker significantly reduced commute distances by 103% and 59%, respectively.

The seemingly unrelated regression showed similar results, and also proved that the error terms between density and distance are negatively correlated.

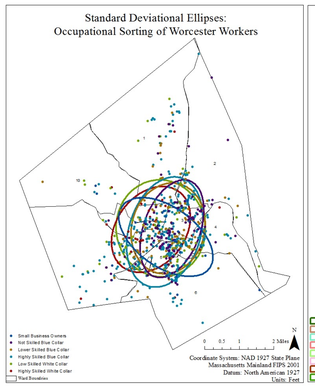
The multivariate logistic regression showed that foreign born individuals and those in an area where their ethnicity is dominant are much more likely to live in high density areas.

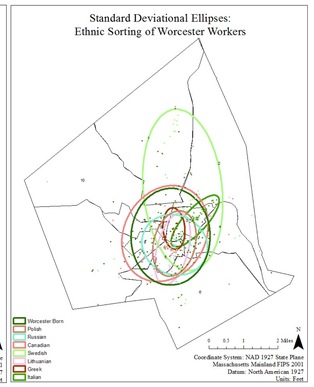
Distance low → high (tertiles)

Space low → high (tertiles)

Skilled Blue Collar: 19% Unskilled Blue Collar: 70% Foreign born: 23% Dominant ethnicity: 322%	Skilled Blue Collar: 2% Unskilled Blue Collar: 24% Foreign born: 121% Dominant ethnicity: 349%	Skilled Blue Collar:150% Unskilled Blue Collar: 200% Foreign born: 357% Dominant ethnicity: 493%
Skilled Blue Collar: 40% Unskilled Blue Collar: 6% Foreign born: 6% Dominant ethnicity: 123%	Skilled Blue Collar: 40% Unskilled Blue Collar: 35% Foreign born: -36% Dominant ethnicity: 174%	Skilled Blue Collar: 82% Unskilled Blue Collar: 133% Foreign born: -30% Dominant ethnicity: 242%
Skilled Blue Collar: 65% Unskilled Blue Collar: 99% Foreign born: 13% Dominant ethnicity: -30%	Skilled Blue Collar: -12% Unskilled Blue Collar: -3% Foreign born: -37% Dominant ethnicity: 26%	Base Case

Blue collar workers are much more likely to live in an area of little space and a long commute to work than an area of lots of space and a long commute to work. Percentages derived from the relative risk ratios of each variable (by choice) are shown in the table. For example, foreign born individuals are 357% more likely than non foreign born individuals to choose high distance, low space, over the base group (high distance, high space)





A grouping analysis in ArcGIS showed that individuals are not sorted by income in the way a monocentric city model would suggest. The additional standard deviation ellipses created show that individuals of similar incomes tend to overlap heavily, but individuals of similar ethnicities are clustered.

Conclusions

The clearest picture created of 1918 Worcester is that of the foreign born individual. Immigrants often lived far from their place of work and lived in extremely high density areas. These individuals were willing to travel further to work when they were able to live near others with similar backgrounds.

The models used in this analysis of Worcester’s urban structure circa 1918 fail to clearly show the concentric circles of income sorting away from the CBD that have been assumed with a monocentric city model. While this study shows that income influences housing decisions, it does not appear to create the model of income sorting that was expected.

Two or three generations from 1918, I would expect ethnic clustering to be much less influential on the spatial structure of the city because immigrant populations had become more “Americanized.” However, with the construction boom from 1900-1920, Worcester was built for the city and the values we find in 1918. If Worcester was built for the values in place in 1918, it has without a doubt, shaped the Worcester of today, even if the values of its inhabitants have changed.